

RM06

Direction Sense

1-2 questions in every RAS Prelims paper without fail, and in some years three. The Pythagoras form (walk East then North, find the shortest distance) is the single most repeated sub-type. Shadow and sun questions come once every two or three papers and are pure rule-recall.

A person walks, turns, walks again, and you are asked how far he ended up from where he started and in which direction. There is no trick to it once you stop drawing arrows freehand and start writing coordinates. This chapter converts every walk into a pair of numbers.

Direction Sense

The compass

- N E S W - four cardinal
- NE SE SW NW - four ordinal
- 45 degrees = one step
- Clockwise ring: N NE E SE S SW W NW

Shortest distance

- $\sqrt{x^2 + y^2}$
- 3-4-5, 5-12-13, 8-15-17
- Equal x and y gives $x \times \text{root } 2$
- Direction from the signs of x and y

Turning

- Right = one step clockwise
- Left = one step anti-clockwise
- Turn around and left/right swap
- 90 = two steps, 180 = four steps

Shadow / sun

- Morning: shadow falls WEST
- Evening: shadow falls EAST
- Noon: no usable shadow
- Two persons face-to-face are opposite

Coordinates

- North +y, South -y
- East +x, West -x
- Add the signed legs
- Net displacement, not path length

Mixed types

- Direction + blood relation
- Relative position of villages
- Return-journey total distance
- Displacement that cancels to zero

What the question asks, and where students throw the mark away

Direction sense gives you a walk. The person starts somewhere, covers a leg, turns, covers another leg, and so on. Then one of two things is asked: how far is he from the start, or in which direction is he from the start. Sometimes both. The walk is deliberately written so that a mental picture goes wrong at the third turn.

- Distance and direction from the starting point after a multi-leg walk.
- Shortest distance using Pythagoras - the single most repeated sub-type.
- Turning problems - left and right turns, and turns given in degrees.
- Shadow and sun problems - sunrise, sunset, which way is the person facing.
- Displacement that cancels, so the person ends up back at the start.
- Direction combined with a blood relation in the same stem.

Three leaks. One: the student adds up all the legs walked and calls that the distance from the start - that is the path length, not the displacement. Two: the student forgets that after turning around, left and right have swapped. Three: the student gets the right number and then names the opposite direction, because he never wrote North at the top of the page. All three vanish if you use coordinates.

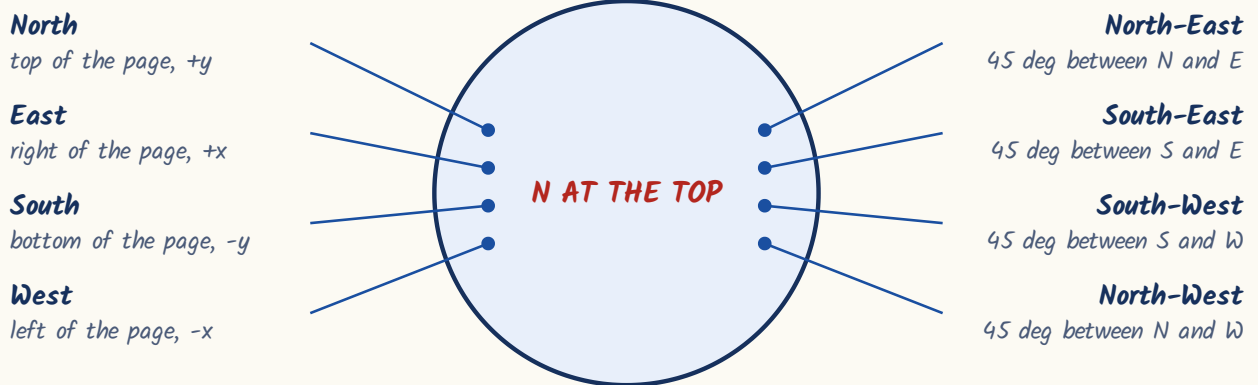
ASKED IN RAS

Direction sense has appeared in every RAS Prelims paper. The turning-walk form came in 2024 (10 km North, right 6 km, right 10 km - answer 6 km), the plain Pythagoras form in 2023 (4 km North then 3 km East - answer 5 km) and 2016 (9 km West then 12 km North - answer 15 km), the cancelling-displacement form in 2021, and the sunrise-shadow form in 2013. Expect at least one, and expect it to be an easy one.

The compass, and the coordinate method that replaces guessing

Before anything else, draw a small compass in the corner of your rough sheet: North at the top, South at the bottom, East to the right, West to the left. Draw it once at the start of the paper and use it for every direction question. This is not decoration - the whole subject depends on the page being oriented the same way each time.

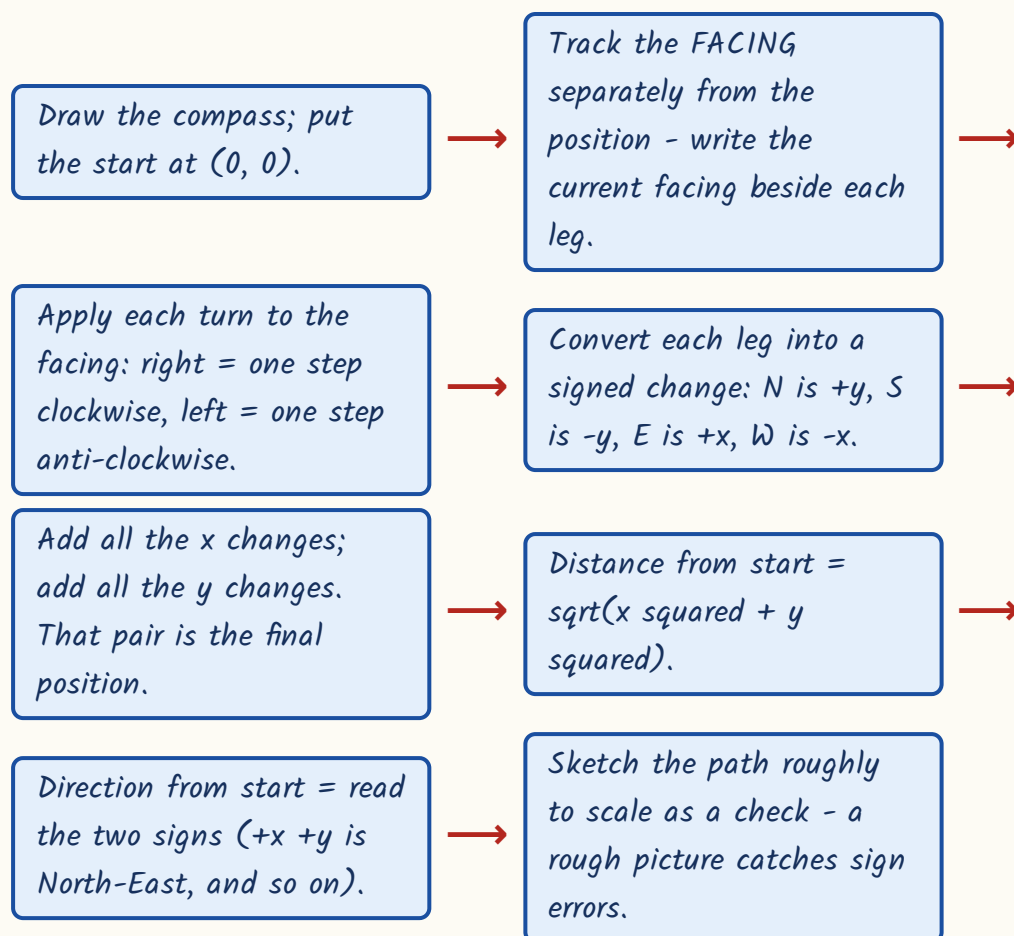
The eight-point compass - draw it once at the top of the sheet



- Four cardinal directions: North, East, South, West.
- Four ordinal directions sit between them: North-East, South-East, South-West, North-West.
- The eight-point ring clockwise: N, NE, E, SE, S, SW, W, NW, back to N.
- One step around that ring is 45 degrees.
- 90 degrees = two steps; 135 degrees = three steps; 180 degrees = four steps.
- Clockwise cardinal order is N to E to S to W; anti-clockwise is the reverse.

Now the method itself. Stop drawing the walk as a wandering line. Convert it into coordinates. Put the starting point at (0, 0). North adds to y, South subtracts from y, East adds to x, West subtracts from x. Add up the signed legs and you have the final position in one line of arithmetic, with no chance of a sign error.

The coordinate method - every walk, every time



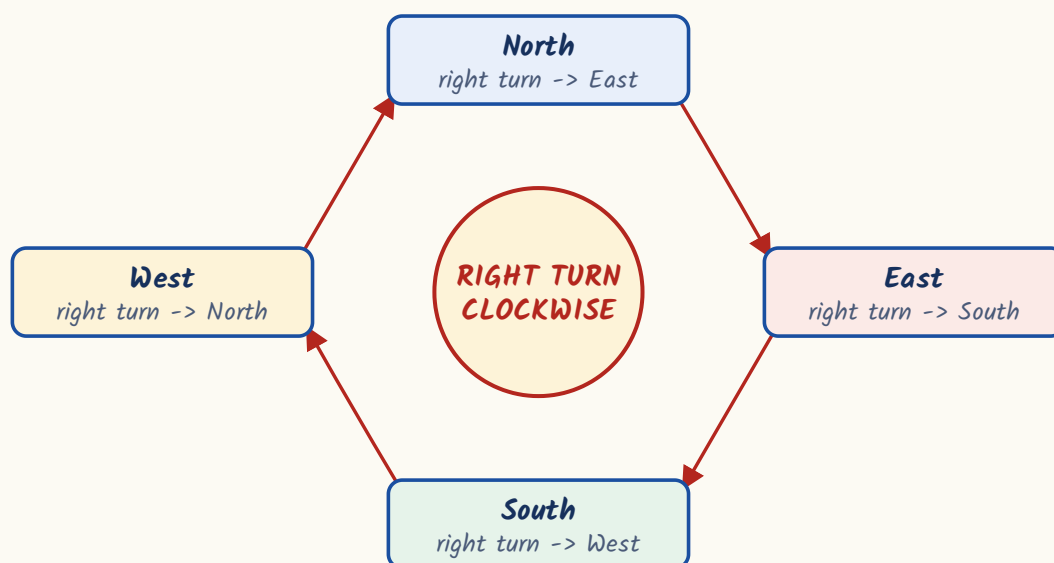
YAAD RAKHO

Remember - NEWS reads clockwise if you start at North and skip: N, E, S, W. A RIGHT turn walks you FORWARD along NEWS; a LEFT turn walks you BACKWARD along it. That one line answers every left-right turn in the paper.

Turning - and the trap that decides more marks than any other

A right turn moves your facing one step clockwise. A left turn moves it one step anti-clockwise. That is all there is to it, but you must apply it to the CURRENT facing, not to North. Write the facing down after every turn; do not carry it in your head.

A right turn walks you one step round the ring



Left and right for each facing - learn this as a block

Facing	Right turn gives	Left turn gives
North	East	West
East	South	North
South	West	East
West	North	South

TRAP

The single most tested trap in the chapter: when the person turns around and faces the **OPPOSITE** direction, his left and his right interchange. Facing North his right is East; turn him to face South and his right is now West. A student who solved the first half of the walk correctly and then forgets this gets the mirror-image answer, which the paper always supplies as an option.

The same person, two facings - notice everything flips

Facing NORTH	Facing SOUTH
Right hand points East	Right hand points West
Left hand points West	Left hand points East
Behind him is South	Behind him is North
A right turn takes him East	A right turn takes him West
Morning shadow falls to his LEFT (West)	Morning shadow falls to his RIGHT (West)
A 45-degree clockwise turn takes him to North-East	A 45-degree clockwise turn takes him to South-West

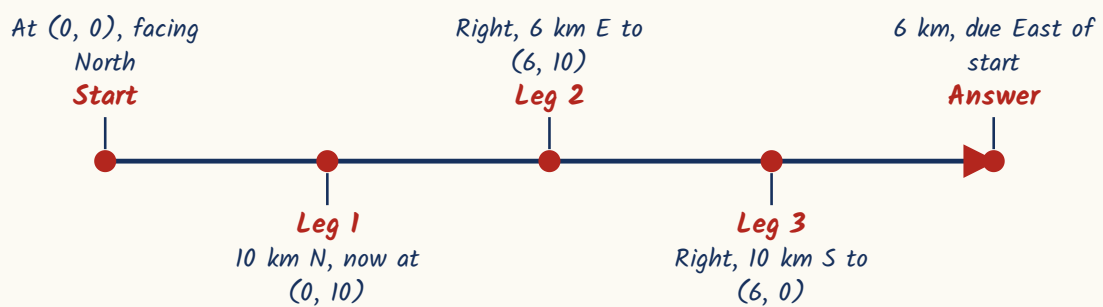
- Turns given in degrees use the eight-point ring: 45 degrees is one step.
- Facing North-East, 90 degrees clockwise = two steps = South-East.
- Facing South-West, 135 degrees clockwise = three steps = North.
- Two persons standing face to face always face opposite directions.
- The direction of A from B is the exact reverse of the direction of B from A - flip both letters, so North-East becomes South-West.

Worked examples - the turning walk

Worked example 1 (asked in RAS Pre 2024). Ramesh walks 10 km towards the North, then turns right and walks 6 km, then turns right again and walks 10 km. How far is he now from his starting point, and in which direction?

- Start at $(0, 0)$, facing North after the first instruction.
- Leg 1: 10 km North. y goes up by 10. Position $(0, 10)$.
- Turn right from North - facing is now East.
- Leg 2: 6 km East. x goes up by 6. Position $(6, 10)$.
- Turn right from East - facing is now South.
- Leg 3: 10 km South. y goes down by 10. Position $(6, 0)$.
- Final position $(6, 0)$. The y parts cancelled completely.
- Distance = 6 km. Direction: x is positive, y is zero, so he is due EAST of the start.

Worked example 1 as coordinates - RAS Pre 2024



Worked example 2. Priya walks 10 km South, turns left and walks 5 km, turns left again and walks 10 km, then turns right and walks 4 km. How far is she from her starting point?

- Start (0, 0), facing South.
- Leg 1: 10 km South. Position (0, -10).
- Turn left from South - one step anti-clockwise gives East. Careful here; facing South reverses the usual sense.
- Leg 2: 5 km East. Position (5, -10).
- Turn left from East - anti-clockwise gives North.
- Leg 3: 10 km North. Position (5, 0).
- Turn right from North - clockwise gives East. Leg 4: 4 km East. Position (9, 0).
- Distance = 9 km, due East. Note the total walked was 29 km - that number is not the answer.

Worked example 3. A man walks 25 km East, then 10 km North, then 15 km West. How far is he from the starting point and in which direction?

- x changes: +25 then -15, giving $x = +10$.
- y changes: +10 only, giving $y = +10$.
- Final position (10, 10).
- x and y are equal, so the distance is 10 times root 2 = about 14.14 km.
- Both signs are positive, so the direction is exactly North-East.
- Whenever x equals y in size, the answer is one of NE, NW, SE, SW and the distance carries a root 2.

YAAD RAKHO

Remember - if the North legs and the South legs add to the same number, and the East legs and the West legs also add to the same number, the person is standing exactly where he started. Distance zero. RPSC sets this as a whole question and the answer 'at the starting point' is always in the options.

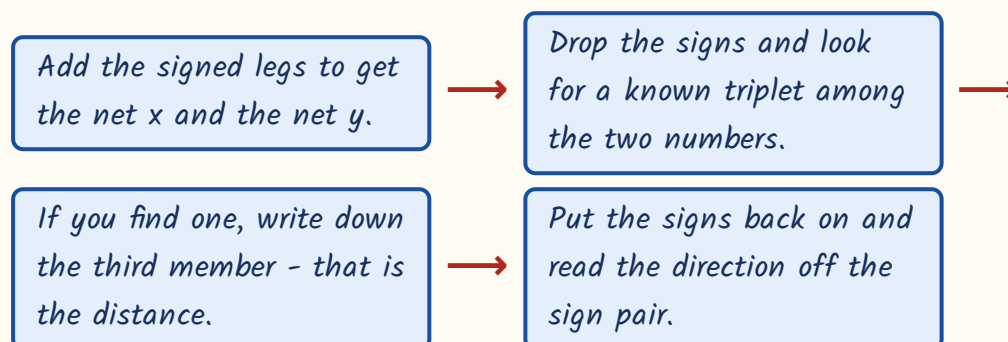
Shortest distance - Pythagoras and the triplets you must know

cold

When the net movement has both an East-West part and a North-South part, the straight line back to the start is the hypotenuse of a right triangle. So the shortest distance is the square root of (x squared plus y squared). The examiner will nearly always pick numbers that make this a whole number, which means he is using a standard Pythagorean triplet. Memorise the triplets and you will never actually compute a square root.

- 3-4-5 and its multiples 6-8-10, 9-12-15, 12-16-20, 15-20-25.
- 5-12-13 - the second most used triplet in RAS papers.
- 8-15-17, 7-24-25, 20-21-29 - the three that catch students out.
- If x and y are equal, distance = x times root 2, about 1.414 times x .
- Signs give the direction: $+x +y$ is NE, $-x +y$ is NW, $+x -y$ is SE, $-x -y$ is SW.
- 'Minimum distance to return' means the straight line back, not the path retraced.

Shortest distance - four steps



Worked example 4 (asked in RAS Pre 2016). A jeep travels 9 km towards the West and then 12 km towards the North. What is the shortest distance from the starting point?

- West 9 km gives $x = -9$. North 12 km gives $y = +12$.
- The two legs are perpendicular, so Pythagoras applies directly.
- Ignore signs: the sides are 9 and 12. That is 3 times (3, 4), so the triplet is 9-12-15.
- Shortest distance = 15 km. No square root needed.
- If the direction had been asked: x negative, y positive, so North-West of the start.

Worked example 5. A bus goes 30 km East and then 40 km North. If it now returns to the starting point by the shortest route, what total distance has it covered in the whole journey?

- Read the question carefully - it asks for the WHOLE journey, not the displacement.
- Outward legs: $30 + 40 = 70$ km.
- Return leg is the hypotenuse of a 30-40 right triangle. That is 10 times (3, 4, 5), so 50 km.
- Whole journey = $70 + 50 = 120$ km.
- The distractor 50 km is the shortest distance alone; the distractor 70 km is the outward path. Both will be in the options.

The shadow rule - sunrise, sunset, and which way you are facing

This sub-type rests on one physical fact you already know. The Sun rises in the East, so in the MORNING every shadow is thrown towards the WEST. The Sun sets in the West, so in the EVENING every shadow is thrown towards the EAST. Once you fix which way the shadow points, the question tells you where the shadow falls relative to the person, and you work backwards to his facing.

Morning (shadow points WEST) - reading the person's facing

Shadow appears	Person is facing
In front of him	West
Behind him	East
To his right	South
To his left	North

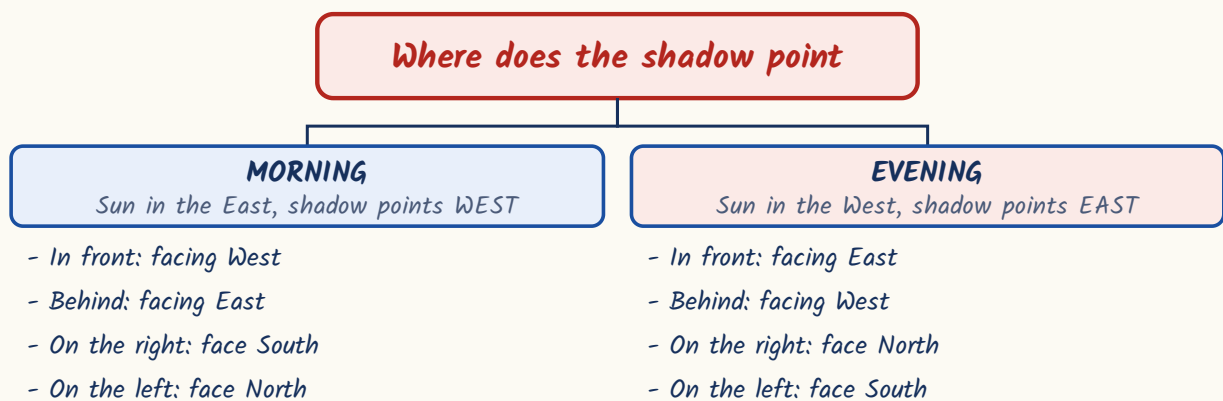
Worked example 6 (asked in RAS Pre 2013). One morning, just after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. Which direction was Suresh facing?

- It is just after sunrise, so the Sun is in the East.
- Every shadow therefore falls towards the West. The pole's shadow points West.
- The stem says that shadow is to Suresh's RIGHT. So West is on his right.
- Rule to use: your facing is 90 degrees ANTI-CLOCKWISE from wherever your right hand points.
- 90 degrees anti-clockwise from West is South. Check it: facing South, right is West. Correct.
- Answer: Suresh was facing South.

TRAP

In the evening every one of the four answers in that table is reversed, because the shadow now points East instead of West. Morning shadow on the right means facing South; evening shadow on the right means facing North. Read the words 'sunrise', 'morning', 'evening' or 'sunset' before you touch the geometry. And at noon there is no usable shadow at all - such a question either says 'no shadow' or is defective.

The shadow rule - fix the time of day before the geometry



Direction combined with blood relations, and the checklist

RPSC sometimes bolts a family relation onto a direction stem: the houses of three people are placed by direction, and the same three people are also related. Do not try to solve both at once. Draw the coordinate diagram, answer the distance part, then draw the family tree separately and answer the relation part. They never interact.

Worked example 7. P's house is 8 km West of Q's house, and R's house is 6 km North of Q's house. P is the brother of Q, and Q is the mother of R. What is the shortest distance between P's house and R's house, and how is P related to R?

- Put Q at the origin (0, 0) because both other houses are described relative to Q.
- P is 8 km West of Q, so P is at (-8, 0).
- R is 6 km North of Q, so R is at (0, 6).
- Difference between P and R: 8 in x and 6 in y. That is 2 times (3, 4), so the triplet is 6-8-10.
- Shortest distance = 10 km.
- Now the separate family tree: Q is R's mother, and P is Q's brother.
- P is the brother of R's mother, so P is R's maternal uncle. Answer: 10 km; maternal uncle.

- Draw the compass first, every time. North at the top.
- Write the facing beside every leg; update it at each turn before moving.
- Convert to coordinates rather than following the line with your eye.
- Answer displacement, not the sum of the legs, unless the words 'total distance covered' appear.
- Look for a triplet before reaching for a square root.
- Read the sign pair to name the direction; never eyeball it.
- Reverse both letters when the question asks for the opposite direction.
- For shadow questions, first fix morning or evening, then place the shadow.

ASKED IN RAS

Asked in RAS Pre 2021 - from a point, a man walks 5 km South, 5 km East, 5 km North and 5 km East; what is his distance and direction from the start? Answer: 10 km East. The South and North legs cancel and the two East legs add. This cancelling form has now appeared in several papers; recognise it in the first reading and it takes fifteen seconds.

REVISION RECAP

1. Draw the compass once at the top of the rough sheet: North up, South down, East right, West left.
2. Eight-point ring clockwise: N, NE, E, SE, S, SW, W, NW. One step = 45 degrees.
3. Right turn = one step clockwise; left turn = one step anti-clockwise, applied to the CURRENT facing.
4. Facing North: right is East. Facing South: right is West. Turning around swaps left and right.
5. 90 degrees is two steps on the ring, 135 degrees is three, 180 degrees is four.
6. Coordinates: North +y, South -y, East +x, West -x. Add the signed legs.
7. Distance from the start is the net displacement, never the sum of the legs walked.
8. If the N-S legs cancel and the E-W legs cancel, the person is back at the start.
9. Shortest distance = $\sqrt{x^2 + y^2}$ when both components exist.
10. Triplets to know: 3-4-5, 6-8-10, 9-12-15, 5-12-13, 8-15-17, 7-24-25, 20-21-29.
11. Equal x and y gives distance x times root 2 and an exactly diagonal direction.
12. Sign pairs: +x +y is NE, -x +y is NW, +x -y is SE, -x -y is SW.
13. Direction of A from B is the exact reverse of the direction of B from A.
14. Morning shadows fall West because the Sun is in the East; evening shadows fall East.
15. Morning: shadow on the right means facing South; in the evening that answer reverses.
16. At noon the shadow is negligible - the answer is usually 'no shadow'.
17. Two persons facing each other face exactly opposite directions.
18. For mixed stems, solve the coordinate diagram and the family tree separately.